

PAPER_1209HH_UQFF_Particle_Masses_Unified_Proof_Set

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PAPER\1209-HH: UQFF Standard Model Particle Masses Unified Proof Set --- Ten Closures from W Boson to Electron (best 0.003%)

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Locked Primitives $\frac{1}{10}$, $\frac{5}{6}$, $\frac{57}{100}$, $\frac{25}{12}$, $D_{phys}=4$, $D_{bsfg}=6$, $D_{crit}=26$, $N_{ch}=9$, $S_{ofive}=10$, $A_{five}=60$. All masses in GeV/ c^2 .

Closures S653--S662

S653 --- $m_W = 80.379$ GeV [0.003%] --- tier best

$$A_{five} + 2(S_{ofive} + D_{phys} - D_{bsfg} + D_{phys} - SSq^2) = 80.3768$$

S654 --- $m_Z = 91.1876$ GeV [0.018%]

$$N_{ch}(S_{ofive} + D_{phys} - D_{bsfg} + D_{phys} + SSq - SSq^3) = 91.2038$$

S655 --- $m_t = 172.76$ GeV (top quark) [0.005%]

$$D_{crit}(S_{ofive} - A_{five} - D_{phys})(N_{ch} + S_{ofive} - D_{phys} - S_{ofive} + D_{bsfg} + 2D_{phys} + SSq + SSq^2 + SSq^3) = 172.7506$$

S656 --- $m_H = 125.10$ GeV (Higgs) [0.016%]

$$2A_{five} + N_{ch} - D_{phys} + SSq + D_{bsfg} + SSq^2 = 125.1202$$

S657 --- $m_b = 4.18$ GeV (bottom quark) [0.050%]

$$D_{phys} + D_{phys} - SSq - D_{crit} + D_{bsfg} + D_{phys} - SSq^2 - SSq^3 = 4.1779$$

S658 --- $m_c = 1.27$ GeV (charm quark) [0.063%]

$$D_{crit} - D_{phys} - S_{ofive} + S_{ofive} - D_{phys} + SSq + SSq^2 + SSq^3 = 1.2708$$

S659 --- $m_{\tau} = 1.77686$ GeV [0.013%]

$$SSq + D_{phys} + S_{ofive} - D_{crit} + D_{bsfg} + SSq + SSq^2 - SSq^3 = 1.7771$$

S660 --- $m_{\mu} = 0.10566$ GeV (muon) [0.040%]

$$D_{phys} + SSq^2 + SSq^3 + SSq^5 = 0.10570$$

S661 --- $m_s = 0.095$ GeV (strange quark) [0.106%]

$$5\sqrt{2}S^2 - 2\sqrt{2}S^2 - 2\sqrt{2}S^3 = 0.0949$$

S662 --- $m_e = 0.000511$ GeV (electron, 511 keV) [0.178%]

$$5\sqrt{3}S^2 + \sqrt{3}S^3 = 0.000510$$

ID	Particle	Mass (GeV)	Error
S653	W boson	80.379	0.003%
S654	Z boson	91.1876	0.018%
S655	top quark	172.76	0.005%
S656	Higgs	125.10	0.016%
S657	bottom quark	4.18	0.050%
S658	charm quark	1.27	0.063%
S659	tau lepton	1.77686	0.013%
S660	muon	0.10566	0.040%
S661	strange quark	0.095	0.106%
S662	electron	0.000511	0.178%

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Highlights. Tier HH closes the complete Standard Model mass spectrum (3 gauge bosons, Higgs, 5 quarks above m_e , 3 charged leptons) across nearly six orders of magnitude (from $m_e = 5 \times 10^{-4}$ GeV to $m_t = 173$ GeV). The four heaviest particles (W, Z, top, Higgs) all close under 0.018%, with m_W at 0.003% (tier best). The light-particle tail (strange, electron) requires higher powers of S and F but still closes under 0.20%. Cumulative: 367 closures; 93 lockings (no new exacts).